

T. Tony Cai
Daniel H. Silberberg Professor
Professor of Statistics and Data Science

Curriculum Vitae

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Education:

- Ph.D. Cornell University, Statistics, 1996

Academic Appointments:

University of Pennsylvania:

- Daniel H. Silberberg Professor, The Wharton School, 2018 – present
- Dorothy Silberberg Professor, The Wharton School, 2007-18
- Assistant-Full Professor of Statistics, The Wharton School, 2000-07
- Professor, Applied Mathematics & Computational Science Graduate Group, 2007 – present
- Associate Scholar, Department of Biostatistics, Epidemiology, & Informatics, Perelman School of Medicine, 2006 – present

Other Positions:

- Visiting Fellow, Isaac Newton Institute for Math. Sciences, Cambridge University, 1 & 6/2008
- Visiting Fellow, Centre for Math. and its Applications, Australian National University, 6-7/2004
- Assistant Professor of Statistics, Purdue University, 1996-2000

Administrative Appointment:

- Vice Dean, The Wharton School, 2017-20

Editorial Appointments:

- Co-Editor, *The Annals of Statistics*, 2010-12
- Associate Editor, *Journal of the Royal Statistical Society, Series B*, 2014-18
- Associate Editor, *Journal of the American Statistical Association*, 2005-10
- Associate Editor, *The Annals of Statistics*, 2004-09
- Associate Editor, *Statistica Sinica*, 2005-11
- Associate Editor, *Statistics Surveys*, 2006-09
- Editorial Board, *Frontiers of Statistics* (book series), 2008 - present
- Guest Editor, *Statistica Sinica* Special Issue on Multiscale Methods
- Guest Editor, *J. Nonparametric Statistics* Special Issue for the Inaugural IMS-China Conference

Awards and Honors:

- Wald Memorial Award and Lecture, Institute of Mathematical Statistics, 2027
- Leo Breiman Senior Award, American Statistical Association, 2026
- Best Paper Award (Gold), International Congress of Chinese Mathematicians, 2026
- President-elect, President, Past President, Institute of Mathematical Statistics, 2023-26
- Fellow, American Association for the Advancement of Science (AAAS), 2024
- Noether Distinguished Scholar Award, American Statistical Association, 2023
- Frontiers of Science Award, International Congress of Basic Science, 2023
- Laplace Lecture of the Bernoulli Society, 10th World Congress in Probability & Statistics, 2021
- International Chinese Statistical Association Distinguished Achievement Award, 2019
- International Chinese Statistical Association Outstanding Service Award, 2019
- Peter Whittle Lecture, Cambridge University, 2018
- Best Paper Award, International Congress of Chinese Mathematicians, 2018
- President-elect, President, Past President, International Chinese Statistical Association, 2016-18
- Hermann Otto Hirschfeld Lectures, Humboldt-Universität zu Berlin, 2012
- Forum Lecture, 28th European Meeting of Statisticians, 2010
- Medallion Lecture, Institute of Mathematical Statistics, 2009
- COPSS Presidents' Award, Committee of the Presidents of Statistical Societies (COPSS), 2008
- Fellow, Institute of Mathematical Statistics, 2006

Research Interests:

- Statistical machine learning
- High-dimensional statistics
- Large-scale inference
- Statistical decision theory
- Functional data analysis & nonparametric function estimation
- Applications to genomics, financial engineering, and medical imaging

Research Grants:

- NSF DMS-2413106, Principal Investigator, 2024-27
Multi-source Learning: Data-driven Algorithms, Optimality Theory, and Applications
- NIH R01-GM129781, Co-Investigator, 2023-2027
Methods for integrative genomic data analysis
- NIH R01-GM123056, Co-Investigator, 2022-26
Statistical methods for microbiome and metagenomics
- NSF DMS-2015259, Principal Investigator, 2020-23
Transfer learning for large-scale inference: General framework and data-driven algorithms
- NIH R01-GM129781, Co-Investigator, 2018-22
Methods for integrative genomic data analysis
- NIH R01-GM123056, Co-Investigator, 2017-21

Statistical methods for microbiome and metagenomics

- NSF DMS-1712735, Principal Investigator, 2017-20
Integrative large-scale data analysis and statistical inference
- Wharton School Global Initiatives, Principal Investigator, 2016-17
A quantitative study of Beijing's air pollution and its economic impacts
- NSF DMS-1403708, Principal Investigator, 2014-17
Theory and methods for estimation of nonsmooth functionals and detection of simultaneous signals
- NIH R01 CA-127334, Co-Investigator, 2012-17
Methods for genomic data with graphical structures
- NSF DMS-1208982, Principal Investigator, 2012-15
Random matrix theory and high dimensional statistics
- NSF FRG DMS-0854973, Principal Investigator, 2009-13
Statistical inference for high-dimensional data: Theory, methodology and applications
- NSF DMS-0604954, Principal Investigator, 2006-09
Theory and methodology for sparse inference
- NSF DMS-0306576, Co-Principal Investigator, 2003-06
An adaptation theory for nonparametric function estimation
- NSF DMS-0072578, Principal Investigator, 2000-03
Block thresholding methods for adaptive wavelet estimation: Theory and applications

Books:

1. Cai, T.T. and Shen, X. (eds.) (2010). *High Dimensional Data Analysis*, World Scientific, New Jersey.
2. Berger, J., Cai, T.T., and Johnstone, I. M. (eds.) (2010). *Borrowing Strength: Theory Powering Applications – A Festschrift for Lawrence D. Brown*. Institute of Mathematical Statistics.

Published/Accepted Papers: (Available at www-stat.wharton.upenn.edu/~tcgai)

- [1] Cai, T.T., Chakraborty, A., and Vuursteen, L. (2026+). Optimal federated learning for nonparametric regression with heterogeneous distributed differential privacy constraints. *Journal of the American Statistical Association*, to appear.
- [2] Cai, T.T., Chakraborty, A., and Wang, Y. (2026+). Optimal differentially private ranking from pairwise comparisons. *Journal of the American Statistical Association*, to appear.
- [3] Liang, Z., Cai, T.T., Sun, W., and Xia, Y. (2026+). A locally adaptive algorithm for multiple testing with network structure. *Statistica Sinica*, to appear.
- [4] Audy, A., Chakraborty, A., and Cai, T.T. (2026). Minimax and adaptive transfer learning for nonparametric classification under distributed differential privacy constraints. *Journal of the Royal Statistical Society, Series B* **88**, 903-929.
- [5] Li, S., Cai, T.T., and Li, H. (2026). Statistical inference for high dimensional regression with proxy data. *Statistica Sinica* **36**, 331-349.

- [6] Cai, T.T., Li, S., and Xiang, D. (2026). Estimation of the unimodal strong stochastic transitivity model and its application in Premier League rankings (in Chinese). *Scientia Sinica Mathematica* **56**, 3-32.
- [7] Chakraborty, A., Auddy, A., and Cai, T.T. (2026). When data can't meet: Estimating correlation across privacy barriers. *Advances in Neural Information Processing Systems* **38**, 95086-95120.
- [8] Rakshit, P., Wang, Z., Cai, T.T., and Guo, Z. (2025). SIHR: Statistical inference in high-dimensional linear and logistic regression models. *The R Journal* **16**, 27-45.
- [9] Tan, J., Zhang, Y., Hong, C., Cai, T.T., Cai, T., and Zhang, A. (2025). Integrated analysis for electronic health records with structured and sporadic missingness. *Journal of Biomedical Informatics* **171**, 104933.
- [10] Wang, S., Yuan, B., Cai, T.T., and Li, H. (2024). Phylogenetic association analysis with conditional rank correlation. *Biometrika* **111**, 881-902.
- [11] Cai, T.T., Ke, Z.T., and Turner, P. (2024). Testing high-dimensional multinomials with applications to text analysis. *Journal of the Royal Statistical Society, Series B* **86**, 922-942.
- [12] Li, S., Zhang, L., Cai, T.T., and Li, H. (2024). Estimation and inference for high-dimensional generalized linear models with knowledge transfer. *Journal of the American Statistical Association* **119**, 1274-1285.
- [13] Cai, T.T. and Wei, H. (2024). Distributed Gaussian mean estimation under communication constraints: Optimal rates and communication-efficient algorithms. *Journal of Machine Learning Research* **25**, 1-63.
- [14] Cai, T.T., Kim, D., and Pu, H. (2024). Transfer learning for functional mean estimation: Phase transition and adaptive algorithms. *The Annals of Statistics* **52**, 654-678.
- [15] Ma, R., Guo, Z., Cai, T.T., and Li, H. (2024). Statistical inference for genetic relatedness based on high-dimensional logistic regression. *Statistica Sinica* **34**, 1023-1043.
- [16] Cai, T.T., Chen, R., and Zhu, Y. (2024). Estimation and inference for minimizer and minimum of convex functions: Optimality, adaptivity, and uncertainty principles. *The Annals of Statistics* **52**, 392-411.
- [17] Cai, C., Cai, T.T., and Li, H. (2024). Transfer learning for contextual multi-armed bandits. *The Annals of Statistics* **52**, 207-232.
- [18] Cai, T.T. and Ma, R. (2024). Matrix reordering for noisy disordered matrices: Optimality and computationally efficient algorithms. *IEEE Transactions on Information Theory* **70**, 509-531.
- [19] Cai, T.T., Guo, Z., and Xia, Y. (2023). Statistical inference and large-scale multiple testing for high-dimensional regression models (invited discussion paper). *Test* **32**, 1135-1171.
- [20] Li, S., Cai, T.T., and Li, H. (2023). Transfer learning in large-scale Gaussian graphical models with false discovery rate control. *Journal of the American Statistical Association* **118**, 2171-2183.
- [21] Wu, R., Zhang, L., and Cai, T.T. (2023). Sparse topic modeling: Computational efficiency, near-optimal algorithms, and statistical inference. *Journal of the American Statistical Association* **118**, 1849-1861.

- [22] Xia, Y. and Cai, T.T. (2023). Discussion of “A Scale-free Approach for False Discovery Rate Control in Generalized Linear Models”. *Journal of the American Statistical Association* **118**, 1569-1572.
- [23] Cai, T.T., Guo, Z., and Ma, R. (2023). Statistical inference for high-dimensional generalized linear models with binary outcomes. *Journal of the American Statistical Association* **118**, 1319-1332.
- [24] Cai, T.T. and Ma, R. (2022). Theoretical foundations of t-SNE for visualizing high-dimensional clustered data. *Journal of Machine Learning Research* **23**, 1-54.
- [25] Li, S., Cai, T.T., and Li, H. (2022). Inference for high-dimensional linear mixed-effects models: A quasi-likelihood approach. *Journal of the American Statistical Association* **117**, 1835-1846.
- [26] Cai, T.T., Zhang, A., and Zhou, Y. (2022). Sparse group Lasso: Sample complexity, optimal rate, and statistical inference. *IEEE Transactions on Information Theory* **68**, 5975-6002.
- [27] Cai, T.T., Sun, W., and Xia, Y. (2022). LAWS: A locally adaptive weighting and screening approach to spatial multiple testing. *Journal of the American Statistical Association* **117**, 1370-1383.
- [28] Li, S., Cai, T.T., and Li, H. (2022). Transfer learning for high-dimensional linear regression: Prediction, estimation, and minimax optimality. *Journal of the Royal Statistical Society, Series B* **84**, 149-173.
- [29] Cai, T.T. and Pu, H. (2022). Stochastic continuum-armed bandits with additive models: Minimax regrets and adaptive algorithm. *The Annals of Statistics* **50**, 2179-2204.
- [30] Cai, T.T. and Wei, H. (2022). Distributed adaptive Gaussian mean estimation with unknown variance: Interactive protocol helps adaptation. *The Annals of Statistics* **50**, 1992-2020.
- [31] Cai, T.T. and Wei, H. (2022). Distributed nonparametric regression: Optimal rate of convergence and cost of adaptation. *The Annals of Statistics* **50**, 698-725.
- [32] Kang, H., Lee, Y., Cai, T.T., and Small, D.S. (2022). Two robust tools for inference about causal effects with invalid instruments. *Biometrics* **78**, 24-34.
- [33] Ma, R., Cai, T.T., and Li, H. (2022). Optimal estimation of simultaneous signals using absolute inner product with applications to integrative genomics. *Statistica Sinica* **32**, 1027-1048.
- [34] Cai, T.T., Han, R., and Zhang, A. (2022). On the non-asymptotic concentration of heteroskedastic Wishart-type matrix. *Electronic Journal of Probability* **27**, 1-40.
- [35] Zhang, A., Cai, T.T., and Wu, Y. (2022). Heteroskedastic PCA: Algorithm, optimality, and applications. *The Annals of Statistics* **50**, 53-80.
- [36] Guo, Z., Renaux, C., Buhlmann, P., and Cai, T.T. (2021). Group inference in high dimensions with applications to hierarchical testing. *Electronic Journal of Statistics* **15**, 6633-6676.
- [37] Cai, T.T., Jiang, T., and Li, X. (2021). Asymptotic analysis for extreme eigenvalues of principal minors of random matrices. *The Annals of Applied Probability* **31**, 2953-2990.
- [38] Cai, T.T., Wang, Y., and Zhang, L. (2021). The cost of privacy: Optimal rates of convergence for parameter estimation with differential privacy. *The Annals of Statistics* **49**, 2825-2850.
- [39] Ma, R., Cai, T.T., and Li, H. (2021). Optimal estimation of bacterial growth rates based on permuted monotone matrix. *Biometrika* **108**, 693-708.

- [40] Cai, T., Cai, T.T., and Guo, Z. (2021). Individualized treatment selection: An optimal hypothesis testing approach in high-dimensional models. *Journal of the Royal Statistical Society, Series B* **83**, 669-719.
- [41] Ma, R., Cai, T.T., and Li, H. (2021). Optimal permutation recovery in permuted monotone matrix model. *Journal of the American Statistical Association* **116**, 1358-1372.
- [42] Wang, S., Cai, T.T., and Li, H. (2021). Optimal estimation of Wasserstein distance on a tree with an application to microbiome studies. *Journal of the American Statistical Association* **116**, 1237-1253.
- [43] Cai, T.T. and Zhang, L. (2021). A convex optimization approach to high-dimensional sparse quadratic discriminant analysis. *The Annals of Statistics* **49**, 1537-1568.
- [44] Wang, S., Cai, T.T., and Li, H. (2021). Hypothesis testing for phylogenetic composition: A minimum-cost flow perspective. *Biometrika* **108**, 17-36.
- [45] Ma, R., Cai, T.T., and Li, H. (2021). Global and simultaneous testing for high-dimensional logistic regression models. *Journal of the American Statistical Association* **116**, 984-998.
- [46] Cai, T.T., Kim, D., Song, X., and Wang, Y. (2021). Optimal sparse eigenspace and low-rank density matrix estimation for quantum systems. *Journal of Statistical Planning and Inference* **213**, 50-71.
- [47] Cai, T.T., Li, H. and Ma, R. (2021). Optimal structured principal subspace estimation: Metric entropy and minimax rates. *Journal of Machine Learning Research* **22**, 1-45.
- [48] Cai, T.T. and Wei, H. (2021). Transfer learning for nonparametric classification: Minimax rate and adaptive classifier. *The Annals of Statistics* **49**, 100-128.
- [49] Xia, Y., Cai, T.T., and Sun, W. (2020). GAP: A general framework for information pooling in two-sample sparse inference. *Journal of the American Statistical Association* **115**, 1236-1250.
- [50] Cai, T.T. and Wu, Y. (2020). Statistical and computational limits for sparse matrix detection. *The Annals of Statistics* **48**, 1593-1614.
- [51] Cai, T.T., Han, X., and Pan, G. (2020). Limiting laws for divergent spiked eigenvalues and largest non-spiked eigenvalue of sample covariance matrices. *The Annals of Statistics* **48**, 1255-1280.
- [52] Cai, T.T. and Guo, Z. (2020). Semi-supervised inference for explained variance in high-dimensional regression and its applications. *Journal of the Royal Statistical Society, Series B* **82**, 391-419.
- [53] Cai, T.T., Liang, T., and Rakhlin, A. (2020). Weighted message passing and minimum energy flow for heterogeneous stochastic block models with side information. *Journal of Machine Learning Research* **21**, 1-34.
- [54] Cai, T.T., Hu, J., Li, Y., and Zheng, X. (2020). High-dimensional minimum variance portfolio estimation based on high-frequency data. *Journal of Econometrics* **214**, 482-494.
- [55] Cai, T.T. (2019). Gaussianization Machines for non-Gaussian function estimation models. *Statistical Science* **34**, 635-656.
- [56] Cai, T.T., Li, H., Ma, J., and Xia, Y. (2019). Differential Markov random field analysis with an application to detecting differential microbial community networks. *Biometrika* **106**, 401-416.

- [57] Xiang, D., Zhao, S. D., and Cai, T.T. (2019). Signal classification for the integrative analysis of multiple sequences of multiple tests. *Journal of the Royal Statistical Society, Series B* **81**, 707-734.
- [58] Cai, T.T. and Zhang, L. (2019). High-dimensional linear discriminant analysis: Optimality, adaptive algorithm, and missing data. *Journal of the Royal Statistical Society, Series B* **81**, 675-705.
- [59] Zhang, A., Brown, L.D., and Cai, T.T. (2019). Semi-supervised inference: General theory and estimation of means. *The Annals of Statistics* **47**, 2538-2566.
- [60] Guo, Z., Wang, W., Cai, T.T., and Li, H. (2019). Optimal estimation of genetic relatedness in high-dimensional linear models. *Journal of the American Statistical Association* **114**, 358-369.
- [61] Cai, T.T., Sun, W., and Wang, W. (2019). CARS: Covariate assisted ranking and screening for large-scale two-sample inference (with discussion). *Journal of the Royal Statistical Society, Series B* **81**, 187-234.
- [62] Cai, T., Cai, T.T., Liao, K., and Liu, W. (2019). Large-scale simultaneous testing of cross-covariance matrix with applications to PheWAS. *Statistica Sinica* **29**, 983-1005.
- [63] Cai, T.T., Ma, J. and Zhang, L. (2019). CHIME: Clustering of high-dimensional Gaussian mixtures with EM algorithm and its optimality. *The Annals of Statistics* **47**, 1234-1267.
- [64] Guo, Z., Kang, H., Cai, T.T., and Small, D.S. (2018). Testing endogeneity with high dimensional covariates. *Journal of Econometrics* **207**, 175-187.
- [65] Basu, P., Cai, T.T., Das, K., and Sun, W. (2018). Weighted false discovery rate control in large-scale multiple testing. *Journal of the American Statistical Association* **113**, 1172-1183.
- [66] Guo, Z., Kang, H., Cai, T.T., and Small, D.S. (2018). Confidence intervals for causal effects with invalid instruments using two-stage hard thresholding with voting. *Journal of the Royal Statistical Society, Series B* **80**, 793-815.
- [67] Cai, T.T., Zhang, L., and Zhou, H.H. (2018). Adaptive functional linear regression via functional principal component analysis and block thresholding. *Statistica Sinica* **28**, 2455-2468.
- [68] Cai, T.T. and Guo, Z. (2018). Accuracy assessment for high-dimensional linear regression. *The Annals of Statistics* **46**, 1807-1836.
- [69] Xia, Y., Cai, T.T., and Li, H. (2018). Joint testing and false discovery rate control for high-dimensional multivariate regression. *Biometrika* **105**, 249-269.
- [70] Xia, Y., Cai, T., and Cai, T.T. (2018). Multiple testing of submatrices of a precision matrix with applications to identification of between pathway interactions. *Journal of the American Statistical Association* **113**, 328-339.
- [71] Cai, T.T., Guntuboyina, A., and Wei, Y. (2018). Adaptive estimation of planar convex sets. *The Annals of Statistics* **46**, 1018-1049.
- [72] Cai, T.T. and Zhang, L. (2018). High-dimensional Gaussian regression: Adaptive estimation and statistical inference. *Statistica Sinica* **28**, 963-993.
- [73] Cai, T.T. and Zhang, A. (2018). Rate-optimal perturbation bounds for singular subspaces with applications to high-dimensional statistics. *The Annals of Statistics* **46**, 60-89.
- [74] Xia, Y., Cai, T., and Cai, T.T. (2018). Two-sample tests for high-dimensional linear regression with an application to detecting interactions. *Statistica Sinica* **28**, 63-92.

- [75] Zhao, S. D., Cai, T.T., and Li, H. (2017). Optimal detection of positive latent dependence between two sequences of multiple tests. *Journal of Multivariate Analysis* **160**, 169-184.
- [76] Cai, T.T. and Sun, W. (2017). Large-scale global and simultaneous inference: Estimation and testing in very high dimensions. *Annual Review of Economics* **9**, 411-439.
- [77] Zhao, S.D., Cai, T.T., Cappola, T.P., Margulies, K.B., and Li, H. (2017). Sparse simultaneous signal detection for identifying genetically controlled disease genes. *Journal of the American Statistical Association* **112**, 1032-1046.
- [78] Cai, T.T., Liang, T., and Rakhlin, A. (2017). On detection and structural reconstruction of small world random networks. *IEEE Transactions on Network Science and Engineering* **4**, 165-176.
- [79] Cai, T.T., Liang, T., and Rakhlin, A. (2017). Computational and statistical boundaries for submatrix localization in a large noisy matrix. *The Annals of Statistics* **45**, 1403-1430.
- [80] Liao, K.P., Sparks, J.A., Hejblum, B., Kuo, I-H., Cui, J., Lahey, L., Cagan, A., Gainer, V.S., Liu, W., Cai, T.T., Sokolove, J., and Cai, T. (2017). Phenome-wide association study of autoantibodies to citrullinated and non-citrullinated epitopes in rheumatoid arthritis. *Arthritis & Rheumatology* **69**, 742-749.
- [81] Cai, T.T. and Guo, Z. (2017). Confidence intervals for high-dimensional linear regression: Minimax rates and adaptivity. *The Annals of Statistics* **45**, 615-646.
- [82] Cai, T.T. and Tan, X.L. (2017). Optimal estimation of a quadratic functional under the Gaussian two-sequence model. *Statistica Sinica* **27**, 879-906.
- [83] Cai, T.T. (2017). Global testing and large-scale multiple testing for high-dimensional covariance structures. *Annual Review of Statistics and Its Application* **4**, 423-446.
- [84] Cai, T.T. and Sun, W. (2017). Optimal screening of sparse signals with applications to multistage high-throughput studies. *Journal of the Royal Statistical Society, Series B* **79**, 197-223.
- [85] Cai, T.T. and Zhang, L. (2016). Discussion of “Influential Feature PCA for High Dimensional Clustering”. *The Annals of Statistics* **44**, 2372-2381.
- [86] Cai, T.T., Li, X. and Ma, Z. (2016). Optimal rates of convergence for noisy sparse phase retrieval via thresholded Wirtinger flow. *The Annals of Statistics* **44**, 2221-2251.
- [87] Cai, T., Cai, T.T., and Zhang, A. (2016). Structured matrix completion with applications to genomic data integration. *Journal of the American Statistical Association* **111**, 621-633.
- [88] Cai, T.T., Liang, T., and Rakhlin, A. (2016). Geometric inference for general high-dimensional linear inverse problems. *The Annals of Statistics* **44**, 1536-1563.
- [89] Cai, T.T. and Zhang, A. (2016). Minimax rate-optimal estimation of high-dimensional covariance matrices with incomplete data. *Journal of Multivariate Analysis* **150**, 55-74.
- [90] Cai, T.T., Eldar, Y.C., and Li, X. (2016). Global testing against sparse alternatives in time-frequency analysis. *The Annals of Statistics* **44**, 1438-1466.
- [91] Cai, T.T. and Zhou, W. (2016). Matrix completion via max-norm constrained optimization. *Electronic Journal of Statistics* **10**, 1493-1525.
- [92] Cai, T.T. and Yuan, M. (2016). Minimax and adaptive estimation of covariance operator for random variables observed on a lattice graph. *Journal of the American Statistical Association* **111**, 253-265.

- [93] Cai, T.T. and Liu, W. (2016). Large-scale multiple testing of correlations. *Journal of the American Statistical Association* **111**, 229-240.
- [94] Kang, H., Zhang, A., Cai, T.T., and Small, D.S. (2016). Instrumental variables estimation with some invalid instruments and its application to Mendelian randomization. *Journal of the American Statistical Association* **111**, 132-144.
- [95] Cai, T.T., Liu, W., and Zhou, H.H. (2016). Estimating sparse precision matrix: Optimal rates of convergence and adaptive estimation. *The Annals of Statistics* **44**, 455-488.
- [96] Cai, T.T., Kim, D., Wang, Y., Yuan, M., and Zhou, H.H. (2016). Optimal large-scale quantum state tomography with Pauli measurements. *The Annals of Statistics* **44**, 681-712.
- [97] Cai, T.T., Li, H., Liu, W., and Xie, J. (2016). Joint estimation of multiple high-dimensional precision matrices. *Statistica Sinica* **26**, 445-464.
- [98] Cai, T.T., Ren, Z., and Zhou, H.H. (2016). Estimating structured high-dimensional covariance and precision matrices: Optimal rates and adaptive estimation (with discussion). *Electronic Journal of Statistics* **10**, 1-59.
- [99] Cai, T.T. and Zhang, A. (2016). Inference for high-dimensional differential matrices. *Journal of Multivariate Analysis* **143**, 107-126.
- [100] Liao, K., Sparks, J.A., Hejblum, B., Kuo, I., Cui, J., Lahey, L.J., Cagan, A., Gainer, V., Liu, W., Cai, T.T. and Sokolove, J. (2015). Phenome-wide association study of novel anti-citrullinated peptide antibodies in rheumatoid arthritis. In *Arthritis & Rheumatology* (Vol. 67), Wiley-Blackwell, New Jersey.
- [101] Cai, T.T. and Li, X. (2015). Robust and computationally feasible community detection algorithm in the presence of outlier nodes. *The Annals of Statistics* **43**, 1027-1059.
- [102] Cai, T.T., Ma, Z., and Wu, Y. (2015). Optimal estimation and rank detection for sparse spiked covariance matrices. *Probability Theory and Related Fields* **161**, 781-815.
- [103] Xia, Y., Cai, T., and Cai, T.T. (2015). Testing differential network with applications to detecting gene by gene interactions. *Biometrika* **102**, 247-266.
- [104] Cai, T.T. and Low, M.G. (2015). A framework for estimation of convex functions. *Statistica Sinica* **25**, 423-456.
- [105] Cai, T.T. and Zhang, A. (2015). ROP: Matrix recovery via rank-one projections. *The Annals of Statistics* **43**, 102-138.
- [106] Jeng, J., Cai, T.T., and Li, H. (2015). Sparse segment identifications with applications to DNA copy number variation analysis. In *Advanced Medical Statistics, 2nd Edition*, Y. Lu, J. Fang, L., Tian, and H. Jin, eds., World Scientific, New Jersey, 863-887.
- [107] Sun, W., Reich, B., Cai, T.T., Guindani, M., and Schwartzman, A. (2015). False discovery control in large-scale spatial multiple testing. *Journal of the Royal Statistical Society, Series B* **77**, 59-83.
- [108] Cai, T.T., Liang, T., and Zhou, H.H. (2015). Law of log determinant of sample covariance matrix and optimal estimation of differential entropy for high-dimensional Gaussian distributions. *Journal of Multivariate Analysis* **137**, 161-172.
- [109] Zhao, S.D., Cai, T.T., and Li, H. (2014). Direct estimation of differential networks. *Biometrika* **101**, 253-268.

- [110] Zhao, S.D., Cai, T.T., and Li, H. (2014). More powerful genetic association testing via a new statistical framework for integrative genomics. *Biometrics* **70**, 881-890.
- [111] Cai, T.T. and Xia, Y. (2014). High-dimensional sparse MANOVA. *Journal of Multivariate Analysis* **131**, 174-196.
- [112] Cai, T.T., Low, M.G., and Ma, Z. (2014). Adaptive confidence bands for nonparametric regression functions. *Journal of the American Statistical Association* **109**, 1054-1070.
- [113] Cai, T.T. and Zhang, A. (2014). Sparse representation of a polytope and recovery of sparse signals and low-rank matrices. *IEEE Transactions on Information Theory* **60**, 122-132.
- [114] Cai, T.T., Liu, W., and Xia, Y. (2014). Two-sample test of high dimensional means under dependence. *Journal of the Royal Statistical Society, Series B* **76**, 349-372.
- [115] Cai, T.T. and Wu, Y. (2014). Optimal detection for sparse mixtures against a given null distribution. *IEEE Transactions on Information Theory* **60**, 2217-2232.
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Service on National/International Committees:

- Executive Committee, Institute of Mathematical Statistics, 2023-26
- Board of Advisors, Institute for Mathematical and Statistical Innovation, 2020-24
- Council of the Institute of Mathematical Statistics, 2020-23
- COPSS Distinguished Achievement Award & Lecture Committee, 2020-23
- Program Chair, IMS Annual Meeting at the Joint Statistical Meetings, Baltimore, 2017
- Publications Committee, Bernoulli Society, 2011-14
- COPSS Presidents' Award Committee, 2009-2012, 2014; Chair, 2012
- Program Committee, Joint Meeting between the IMS and Australian Statistical Conference, 2014
- Council of the Institute of Mathematical Statistics, 2010-12
- Program Committee, 8th World Congress in Probability and Statistics, Istanbul, Turkey, 2012
- Committee on Special Lectures, Institute of Mathematical Statistics, 2008-12
- Director of the Board of International Chinese Statistical Association, 2009-11
- Committee on Nominations, Institute of Mathematical Statistics, 2008-09
- Co-Chair, Scientific Committee, IMS-China International Conference, Weihai, China, 2009
- Scientific Committee, IMS-China International Conference, Hangzhou, China, 2008
- Program Chair, IMS Annual Meeting, Salt Lake City, 2007
- Committee to Select Editors, Institute of Mathematical Statistics, 2000-03
- National Science Foundation CAREER Award Panelist
- National Science Foundation Panelist

Professional Service:

Reviewer for National Science Foundation, National Security Agency, Israel Science Foundation.
Referee for the Annals of Statistics, Journal of the American Statistical Association, Journal of the Royal Statistical Society, Series B, Biometrika, Bernoulli, Probability Theory and Related Fields, Applied and Computational Harmonic Analysis, Statistica Sinica, Scandinavian Journal of Statistics, Journal of Computational and Graphical Statistics, The American Statistician, Journal of Statistical Planning and Inference, Statistics and Probability Letters, Statistical Science, Psychometrika, Science in China, IEEE Transactions on Information Theory, IEEE Transactions on Image Processing, IEEE Transactions on Signal Processing, Journal of Multivariate Analysis, Proceedings of the National Academy of Sciences, Journal of Machine Learning Research.

University and Departmental Service:

Lawrence D. Brown Distinguished Lecture and Ph.D. Student Workshop Committee, 2021-2024
Wharton Faculty IT Steering Committee, 2022-23
Vice Dean for China Initiatives, The Wharton School, 2017-20
Hiring Committee, Department of Statistics, 2017-18
Wharton Dean's Advisory Council, 2016-17
Executive Committee, Applied Math and Computational Science Grad. Group, 2015-17
Wharton Faculty Survey Committee, 2013-14

Hiring Committee, Department of Statistics, 2012-14
 Quinquennial Departmental Review Committee, 2010-11
 Admission Committee, Applied Math and Computational Science Grad. Group, 2008-20
 Doctoral Program Executive Committee, The Wharton School, 2005-08
 Co-director for the Ph.D. Program in Statistics, 2005-08
 Board for the Student Disciplinary System, University of Pennsylvania, 2002-04
 Board for the Code of Academic Integrity, University of Pennsylvania, 2002-04
 Colloquium Chair, Department of Statistics, Spring, 2002
 Computing Committee, Department of Statistics, 2001-02
 Design of Departmental Website, Spring, 2002

Professional Society Membership:

Institute of Mathematical Statistics
 American Statistical Association
 International Chinese Statistical Association
 American Association for the Advancement of Science

Special Invited Lectures:

1. Plenary Lecture, International Congress of Chinese Mathematicians, Shanghai, 1/2026
2. Plenary Lecture, Zhejiang University Symposium on Probability and Statistics, Hangzhou, 12/2025
3. Plenary Lecture, IMS New Researchers Conference, Nashville, 8/2025
4. Plenary Lecture, International Congress of Basic Science, Beijing, 7/2025
5. Plenary Lecture, International Conference on Statistical Optimization and Learning, Beijing, 7/2025
6. Plenary Lecture, International Congress of Basic Science, Beijing, 7/2024
7. Bahadur Lecture, Conference on Theory and Foundations of Statistics in the Era of Big Data, Tallahassee, 4/2024
8. Distinguished Scientist Hall Lecture, Institute of Systems Science, Chinese Academy of Sciences Chinese Academy of Sciences, Beijing, 10/2023
9. Gottfried E. Noether Award Lecture of the American Statistical Association, Joint Statistical Meetings, Toronto, 2023
10. Frontiers of Science Award Lecture, International Congress of Basic Science, Beijing, 7/2023
11. Keynote Lecture, Joint Conference of Statistics and Data Science in China, Beijing, 7/2023
12. Plenary Lecture, International Conference on High-dimensional Statistics, Changsha, 5/2023
13. Distinguished Lecture, Institute for Advanced Study, Hong Kong University of Science and Technology, 5/2023
14. AMA Distinguished Lecture, Hong Kong Polytechnic University, 5/2023
15. Herman Rubin Distinguished Lecture, Purdue University, 4/2023
16. Distinguished Lecture, National University of Singapore, 5/2022
17. Plenary Lecture, 2021 Shenzhen International Conference on Frontiers of Statistics and Data Science, 12/2021

18. Laplace Lecture of the Bernoulli Society, 10th World Congress in Probability & Statistics, 7/2021
19. Plenary Lecture, The 3rd CSIAM Conference on Big Data and Artificial Intelligence, 7/2021
20. Keynote Lecture, 4th Symposium of the ICSA-Canada Chapter, Queen's University in Kingston, 8/2019
21. Keynote Lecture, 12th Annual International Conference of the Chinese Scholars Association for Management Science and Engineering, Chengdu, 7/2019
22. Keynote Lecture, 2019 Lilac International Conference on Application of Statistics, 6/2019
23. Distinguished Lecture, Zhejiang Gongshang University, 5/2019
24. Plenary Lecture, DAMO Academy Workshop on Extremely Large-Scale Graph and Graph Representation in Practice, Hangzhou, 3/2019
25. Joe Talk, 53rd Wharton Global Forum, Shanghai, 3/2019
26. Plenary Lecture, 2019 ICSA Conference on Data Science, 1/2019
27. Distinguished Lecture, 2018 International Conference on Data Science, Shanghai, 12/2018
28. Peter Whittle Lecture, Cambridge University, 10/2018
29. Keynote Lecture, 5th IBS-China Joint Biostatistics Conference, Guangzhou, China, 7/2018
30. Plenary Lecture, 9th International Purdue Symposium on Statistics, 6/2018
31. Keynote Lecture, Computational and Statistical Interface to Big Data Conference, King Abdullah University of Science and Technology (KAUST), Saudi Arabia, 3/2018
32. Distinguished Lecture, Department of Statistics, Chinese University of Hong Kong, 1/2018
33. JRI Distinguished lecture, Hong Kong Polytechnic University, 1/2018
34. Special Invited Lecture, 6th IMS-China Conference, Nanning, 7/2017
35. Distinguished Lecture, Chongqing Normal University, 6/2017
36. Keynote Lecture, 5th Workshop on Biostatistics and Bioinformatics, Atlanta, 5/2017
37. Distinguished Lecture, Southern University of Science and Technology, 3/2017
38. JRI Distinguished lecture, Hong Kong Polytechnic University, 1/2017
39. Keynote Lecture, 10th ICSA International Conference, Shanghai, 12/2016
40. Plenary Lecture, International Conference on "Some Mathematical Approximation Approaches in Data Science", Hangzhou, 12/2016
41. Distinguished Lecture, Institute for Advanced Study & Department of ISOM, Hong Kong University of Science and Technology, 6/2016
42. Keynote Lecture, Symposium: Frontiers of Statistics and Data Science, Hong Kong, 6/2016
43. Keynote Lecture, Workshop on High-Dimensional Statistical Modeling and Analysis in the Era of Big Data, Beijing, 5/2016
44. The Annals of Statistics Editors' special invited talk at the Joint Statistical Meetings, Seattle, 8/2015
45. Keynote Lecture, 2nd International Workshop on Frontiers of Statistics with Applications to Finance, Shanghai, 7/2015
46. Plenary Lecture, 2015 International Workshop on Mathematical Issues in Information Sciences, Xi'An, 7/2015
47. Plenary Lecture, 10th International Conference on Frontiers of Statistics, Beijing, 6/2015
48. Plenary Lecture, International Conference on Robust Statistics, Kolkata, India, 1/2015

49. Distinguished Lecture, Department of Statistics, Chinese University of Hong Kong, 1/2015
50. Distinguished Lecture, Department of Mathematics, Hong Kong Baptist University, 1/2015
51. Plenary Lecture, Workshop on “Statistics and Computational Interface to Big Data”, Institute for Advanced Studies, Hong Kong University of Science and Technology, 1/2015
52. Plenary Lecture, International Conference on Learning and Approximation, Shanghai, 12/2014
53. Keynote Lecture, The 2014 ICSA Shanghai Statistics Conference, Shanghai, 7/2014
54. Distinguished Lecture, Department of Mathematics, Zhejiang University, 6/2014
55. Keynote Lecture, Conference on Big Data and Statistics, Tianjin, 4/2014
56. Special Invited Lecture, 4th IMS-China Conference, Chengdu, 7/2013
57. Keynote Lecture, International Conference on Statistics and Optimization, Beijing, 6/2013
58. Plenary Lecture, 7th Annual Information Theory and Applications Conference, San Diego, 2/2013
59. Distinguished Lecture, Department of Statistics, University of Washington, 2/2013
60. Hermann Otto Hirschfeld Lectures, Humboldt-Universität zu Berlin, 10/2012
61. Distinguished Lecture, 2nd Institute of Mathematical Statistics - Asia Pacific Rim Meeting, Tokyo, 7/2012
62. Distinguished Lecture, Department of Mathematics, Hong Kong Baptist University, 1/2012
63. Plenary Talk, International Research Symposium on Frontiers of Statistics, Hefei, 7/2011
64. Distinguished Lecture, 3rd IMS-China Conference, Xi’An, 7/2011
65. Plenary Lecture, International Conference on High Dimensional Statistics: Advances and Challenges, Singapore, 5/2011
66. Plenary Talk, 4th Lehmann Symposium, Houston, 5/2011
67. Puri Lecture, Department of Statistics, Purdue University, 4/2011
68. Forum Lecture, 28th European Meeting of Statisticians, Piraeus, Greece, 8/2010
69. Plenary Lecture, International Conference on Statistical Analysis of Complex Data, Kunming, 7/2010
70. Keynote Lecture, 9th German Open Conference on Probability and Statistics, Leipzig, 3/2010
71. Medallion Lecture, Institute of Mathematical Statistics Annual Meeting, Washington DC, 8/2009
72. Distinguished Lecture, 1st Institute of Mathematical Statistics - Asia Pacific Rim Meeting, Seoul, 6/2009
73. Keynote Lecture, Symposium on New Directions in Asymptotic Statistics, Athens, 5/2009